# HU Collection

Engineering and Science Admissions Test

# Mock Set F

Chemistry

Time allowed: 40 minutes
Questions: 27 multiple choice
Calculators: not permitted
Written by: Kevin Hu
Date written: 29 October 2026

Name: Date sat: Score: / 27

## Instructions

Answer all 27 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers, so answer every question. Calculators are not permitted. Relative atomic masses and any other data you need are given in the questions. This paper is deliberately longer than the time allows: move on from any question that is costing you more than two minutes, and come back to it at the end.

| **Q** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Ans** | | | | | | | | | | |

| **Q** | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Ans** | | | | | | | | | | |

| **Q** | 21 | 22 | 23 | 24 | 25 | 26 | 27 | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Ans** | | | | | | | | | | |



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**1.** Liquid nitrogen at its boiling point has a density of 0.84 g $cm^{-3}$. A sample of it is allowed to boil, and the gas is then warmed to room temperature and pressure, where one mole of any gas occupies 24 $dm^3$.

Assume that in the liquid and in the gas the molecules are spread evenly, each molecule having an equal share of the volume.

By approximately what factor is the average distance between neighbouring molecules in the gas at room temperature and pressure greater than the average distance between neighbouring molecules in the liquid?

(Relative atomic mass: N = 14)

**A** 1440 **B** 720 **C** 9

**D** 27 **E** 11 **F** 38

**2.** A student investigates the reaction of zinc with acids. In each experiment a large excess of zinc granules is completely covered by the acid, and the hydrogen given off is collected at room temperature and pressure until the reaction stops. From each graph of volume against time she works out the rate of gas production at many moments. She then plots this rate against $V$, the volume of hydrogen collected so far (not against time).

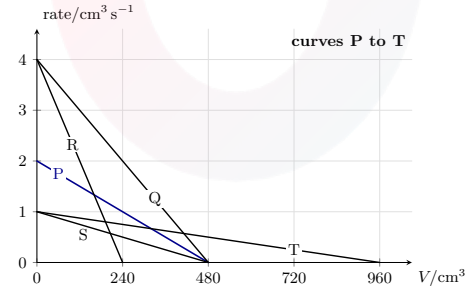
Curve P is for 40.0 $cm^3$ of 1.00 mol $dm^{-3}$ hydrochloric acid, and it is drawn in both panels below for comparison. She then carries out three more experiments with the same zinc at the same temperature:

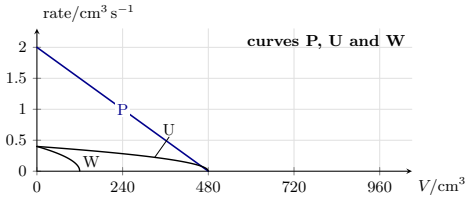
Experiment X: 80.0 $cm^3$ of 0.500 mol $dm^{-3}$ hydrochloric acid.

Experiment Y: 40.0 $cm^3$ of 1.00 mol $dm^{-3}$ ethanoic acid.

Experiment Z: 20.0 $cm^3$ of 2.00 mol $dm^{-3}$ hydrochloric acid.

Assume that the zinc is in such large excess that its surface area stays effectively constant, and that the temperature stays constant.




Which curves show experiments X, Y and Z?

**A** X: T, Y: U, Z: Q

**B** X: S, Y: W, Z: R

**C** X: T, Y: W, Z: R



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**D** X: S, Y: U, Z: Q

**E** X: T, Y: U, Z: R

**F** X: S, Y: P, Z: Q

**G** X: S, Y: U, Z: R

**H** X: S, Y: W, Z: Q

**3.** Each of the following reactions takes place in the treatment of drinking water, in the formation of air pollutants, or in pollution control.

I. Chlorination of water: $Cl_2(aq) + H_2O(l) \rightleftharpoons HCl(aq) + HClO(aq)$

II. Formation of fine particles in smog: $NH_3(g) + HNO_3(g) \rightarrow NH_4NO_3(s)$

III. In a hot car engine: $N_2(g) + O_2(g) \rightarrow 2NO(g)$

IV. Formation of acid rain: $3NO_2(g) + H_2O(l) \rightarrow 2HNO_3(aq) + NO(g)$

V. Removal of sulfur dioxide from flue gases: $CaCO_3(s) + SO_2(g) \rightarrow CaSO_3(s) + CO_2(g)$

VI. Breakdown of ozone: $2O_3(g) \rightarrow 3O_2(g)$

Which of these are redox reactions?

**A** III only

**B** I, III and IV only

**C** III and IV only

**D** I, II, III and IV only

**E** I, III, IV and V only

**F** III, V and VI only

**G** I, III, IV and VI only

**H** I, III, IV, V and VI only

**4.** Prussian blue is a deep blue pigment that has been used by artists for three centuries. Treat it as an ionic solid that contains only two kinds of ion: iron(III) ions, $Fe^{3+}$, and hexacyanoferrate(II) ions.

A hexacyanoferrate(II) ion consists of a single iron atom, in the +2 oxidation state, surrounded by six cyanide groups. Each cyanide group counts as a cyanide ion, $CN^-$, containing one carbon atom and one nitrogen atom. The charge on the whole hexacyanoferrate(II) ion is the total of the charges on its iron and its six cyanide groups.

In Prussian blue, what is the ratio of the total number of iron atoms to the number of nitrogen atoms?

**A** 5 : 18 **B** 1 : 3 **C** 5 : 12

**D** 7 : 24 **E** 2 : 9 **F** 7 : 18



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**5.** Titanium is extracted from rutile, $TiO_2$, in two stages.

Stage 1, at about 900 °C: $TiO_2 + 2C + 2Cl_2 \rightarrow TiCl_4 + 2CO$

Stage 2, under an atmosphere of argon: $TiCl_4 + 2Mg \rightarrow Ti + 2MgCl_2$

The magnesium chloride formed is then electrolysed in the molten state to recover magnesium and chlorine for reuse.

Relative atomic masses: Mg = 24, Ti = 48.

Consider the following statements.

I. Titanium is reduced in stage 1.

II. In stage 1, carbon acts as a reducing agent and chlorine acts as an oxidising agent.

III. In stage 2, titanium(IV) chloride acts as an oxidising agent.

IV. The mass of magnesium that reacts in stage 2 is equal to the mass of titanium produced.

V. In the electrolysis of molten magnesium chloride, magnesium is formed at the positive electrode.

Which of the statements are correct?

**A** I, II, III and IV only

**B** III and IV only

**C** II, III, IV and V only

**D** II, III and IV only

**E** II and IV only

**F** I, II and IV only

**G** I and IV only

**H** II and III only

**6.** A fertiliser contains ammonium nitrate, $NH_4NO_3$, and potassium chloride only. A 3.00 g sample is warmed with excess sodium hydroxide solution. The ammonium ions react to release ammonia,

$NH_4^+(aq) + OH^-(aq) \rightarrow NH_3(g) + H_2O(l)$

while the nitrate ions do not react. All of the ammonia is driven off and absorbed in 50.0 $cm^3$ of 0.500 mol $dm^{-3}$ sulfuric acid, where

$2NH_3(g) + H_2SO_4(aq) \rightarrow (NH_4)_2SO_4(aq)$

The sulfuric acid left over is then titrated with 1.00 mol $dm^{-3}$ sodium hydroxide solution, using an indicator that changes colour as soon as all of the leftover acid has been neutralised. The titre is 20.0 $cm^3$.

What is the percentage by mass of nitrogen in the fertiliser?

(Relative atomic masses: H = 1, N = 14, O = 16)

**A** 9.33% **B** 28.0% **C** 14.0%

**D** 80.0% **E** 4.67% **F** 46.7%



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**7.** Consider a 1.00 g sample of each of the substances below, and count the total number of covalent bonds in each sample. Every covalent bond in these substances is a single bond, and forces between molecules are not counted.

In cubic boron nitride, every boron atom is covalently bonded to four nitrogen atoms and every nitrogen atom is covalently bonded to four boron atoms. White phosphorus consists of $P_4$ molecules in which each phosphorus atom is bonded to the other three. Poly(ethene) consists of very long chains; ignore the atoms at the ends of the chains.

Which sample contains the greatest number of covalent bonds?

(Relative atomic masses: H = 1, B = 11, C = 12, N = 14, O = 16, Si = 28, P = 31)

**A** diamond **B** methane **C** cubic boron nitride

**D** poly(ethene) **E** white phosphorus **F** silicon dioxide

**8.** Hydrogen peroxide, H−O−O−H, decomposes when a little manganese(IV) oxide catalyst is added:

$2H_2O_2 \rightarrow 2H_2O + O_2$

Some catalyst is added to 100 $cm^3$ of hydrogen peroxide solution in an insulated cup. When the decomposition is complete, the temperature of the solution has risen by 5.0 K.

Bond energies: O−O 144 kJ $mol^{-1}$; O=O 498 kJ $mol^{-1}$.

Assume that the enthalpy change of the reaction is the value calculated from bond energies, and that all of the energy released heats the solution, which has mass 100 g and specific heat capacity 4.2 J $g^{-1}$ $K^{-1}$. One mole of any gas occupies 24 $dm^3$ at room temperature and pressure.

What volume of oxygen, measured at room temperature and pressure, is released?

**A** 120 $cm^3$ **B** 71 $cm^3$ **C** 240 $cm^3$

**D** 142 $cm^3$ **E** 240 $dm^3$ **F** 480 $cm^3$

**9.** Compound X is a saturated, non-cyclic alcohol containing only carbon, hydrogen and oxygen. Every oxygen atom in X is part of an OH group, and no carbon atom carries more than one OH group.

When 1.86 g of X is burned completely, 2.64 g of carbon dioxide is formed. When 1.86 g of X reacts with excess sodium, 720 $cm^3$ of hydrogen is produced, measured at room temperature and pressure.

What volume of oxygen, measured at room temperature and pressure, is needed to burn 1.86 g of X completely?

Relative atomic masses: H = 1, C = 12, O = 16. One mole of any gas occupies 24 $dm^3$ at room temperature and pressure.

**A** 2.52 $dm^3$ **B** 1.68 $dm^3$ **C** 3.60 $dm^3$

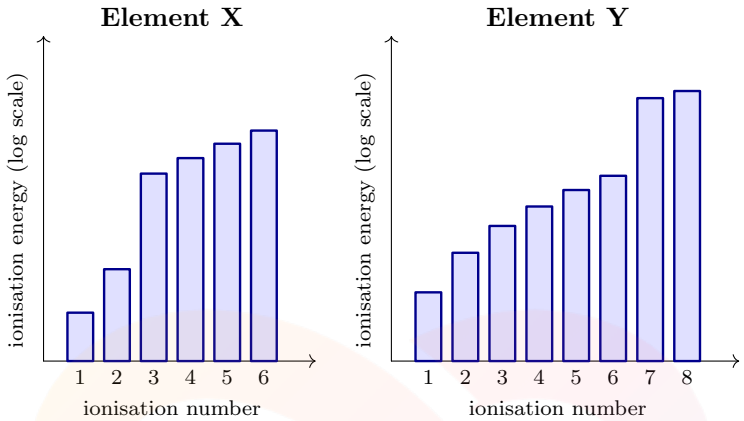
**D** 2.16 $dm^3$ **E** 1.44 $dm^3$ **F** 1.80 $dm^3$



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**10.** The *first ionisation energy* of an element is the energy needed to remove one electron from each atom in a mole of gaseous atoms. The *second ionisation energy* is the energy needed to remove one more electron from each of the resulting 1+ ions, and so on. Successive ionisation energies always increase, but there is a much larger increase than usual whenever the next electron has to be removed from a shell closer to the nucleus than the shell that the previous electron came from.

X and Y are two of the first 20 elements. The graphs show the first six successive ionisation energies of X and the first eight successive ionisation energies of Y, each on a logarithmic scale.




Which of the following statements must be true?

I. An atom of X has electrons in at least three shells.

II. X and Y are in the same period of the Periodic Table.

III. X and Y form an ionic compound with the formula XY.

**A** II and III only

**B** none of them

**C** I and II only

**D** III only

**E** I and III only

**F** I, II and III

**G** I only

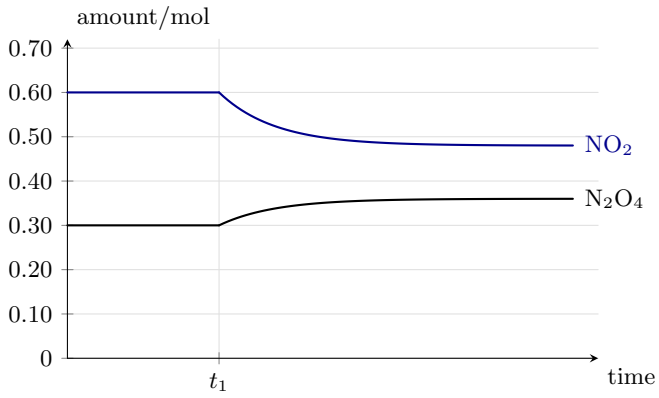


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**11.** Dinitrogen tetroxide and nitrogen dioxide are in dynamic equilibrium in a sealed container:

$N_2O_4(g) \rightleftharpoons 2NO_2(g) \qquad \Delta H = +58 \text{ kJ mol}^{-1}$

At time $t_1$ a single change is made. The graph shows the amount, in moles, of each gas in the container before and after the change.




Which of the following changes, made at time $t_1$, could explain the graph?

I. Decreasing the volume of the container, keeping the temperature constant.

II. Lowering the temperature, keeping the volume of the container constant.

III. Adding a catalyst, keeping the temperature and the volume of the container constant.

**A** I and II only

**B** I only

**C** I and III only

**D** none of them

**E** I, II and III

**F** III only

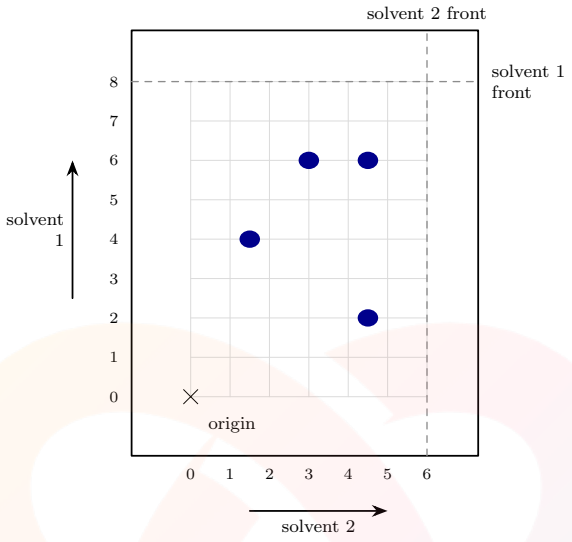
**G** II only

**H** II and III only



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**12.** Two-dimensional paper chromatography can separate mixtures that a single solvent cannot. A spot of a food-colouring mixture is placed at the origin, near one corner of a sheet of chromatography paper. The paper is first developed with solvent 1, which rises up the paper in the direction shown. The paper is then dried, turned through 90° and developed with solvent 2, so that every spot moves on at right angles to its first direction. The positions of both solvent fronts are marked. The finished chromatogram is shown; the grid lines are 1.0 cm apart and the numbers give distances from the origin in cm.




For each solvent, the $R_f$ value of a substance is the distance it moved in that solvent's direction divided by the distance moved by that solvent, both measured from the origin. The table gives the $R_f$ values of six reference dyes, P to U.

| dye | P | Q | R | S | T | U |
|---|---|---|---|---|---|---|
| $R_f$ in solvent 1 | 0.50 | 0.25 | 0.50 | 0.75 | 0.25 | 0.40 |
| $R_f$ in solvent 2 | 0.50 | 0.75 | 0.25 | 0.75 | 0.50 | 0.75 |

Which reference dyes are present in the mixture, and what is the smallest number of different substances that the mixture could contain?

**A** P, Q, R, S and T only; at least 5 substances

**B** Q, R and S only; at least 3 substances

**C** P, Q, R and S only; at least 4 substances

**D** Q, R, S and U only; at least 4 substances

**E** Q, R and S only; at least 4 substances

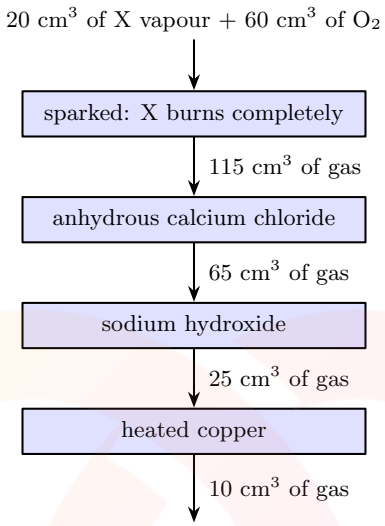
**F** Q, R, S and T only; at least 4 substances



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**13.** Compound X contains only carbon, hydrogen, nitrogen and oxygen. A sample of X vapour is mixed with excess oxygen and sparked, so that X burns completely to carbon dioxide, water and nitrogen, $N_2$, only. The gas is then passed in turn through three absorbers: anhydrous calcium chloride, which removes only water; sodium hydroxide, which removes only carbon dioxide; and heated copper, which removes only oxygen.

The diagram shows the volume of gas at each stage. All volumes are measured at the same temperature and pressure, at which X and water are both gases.




What is the molecular formula of X?

**A** $C_4H_{10}NO_4$ **B** $C_2H_5NO_2$ **C** $C_2H_5NO_3$

**D** $C_4H_5N_2O_4$ **E** $C_2H_5N$ **F** $C_4H_{10}N_2O$

**14.** A geologist tests a sample of crushed limestone, which is calcium carbonate mixed with silica. Silica does not react with acid. She adds limestone to hydrochloric acid of unknown concentration, waits until the reaction has stopped, and measures the total volume of carbon dioxide given off at room temperature and pressure:

$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

Experiment 1: 2.00 g of limestone with 50.0 $cm^3$ of the acid gives 360 $cm^3$ of carbon dioxide.

Experiment 2: 4.00 g of limestone with 50.0 $cm^3$ of the same acid gives 480 $cm^3$ of carbon dioxide.

What volume of carbon dioxide is given off when 3.00 g of the same limestone is added to 40.0 $cm^3$ of the same acid?

(Relative atomic masses: C = 12, O = 16, Ca = 40. One mole of any gas occupies 24 $dm^3$ at room temperature and pressure. Assume that no carbon dioxide dissolves.)

**A** 288 $cm^3$ **B** 480 $cm^3$ **C** 360 $cm^3$

**D** 540 $cm^3$ **E** 384 $cm^3$ **F** 768 $cm^3$



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**15.** A pale green crystalline solid X is a single compound. A student carries out the following tests.

I. Gentle heating of X in a dry tube gives a colourless liquid on the cooler part of the tube, which turns white anhydrous copper(II) sulfate blue.

II. X gives no characteristic flame colour.

III. Sodium hydroxide solution added to a solution of X gives a green precipitate, whose surface slowly turns orange-brown on standing in air. On warming the mixture, a gas that turns damp red litmus paper blue is given off; this gas is ammonia, formed from ammonium ions.

IV. A solution of X acidified with dilute hydrochloric acid gives a white precipitate with barium chloride solution.

V. A solution of X acidified with dilute nitric acid gives no precipitate with silver nitrate solution.

X contains only the ions identified by these tests, together with water of crystallisation.

Next, 7.84 g of X is dissolved in water and made up to 250.0 $cm^3$ of solution. Excess barium chloride solution is added to a 25.0 $cm^3$ portion, and the precipitate is filtered off, washed and dried: its mass is 0.932 g. A second 25.0 $cm^3$ portion is warmed with excess sodium hydroxide solution until all the ammonia has been driven off: 96 $cm^3$ of ammonia is collected, measured at room temperature and pressure.

How many molecules of water of crystallisation does X contain for each iron ion?

Relative atomic masses: H = 1, N = 14, O = 16, S = 32, Fe = 56, Ba = 137. One mole of any gas occupies 24 $dm^3$ at room temperature and pressure.

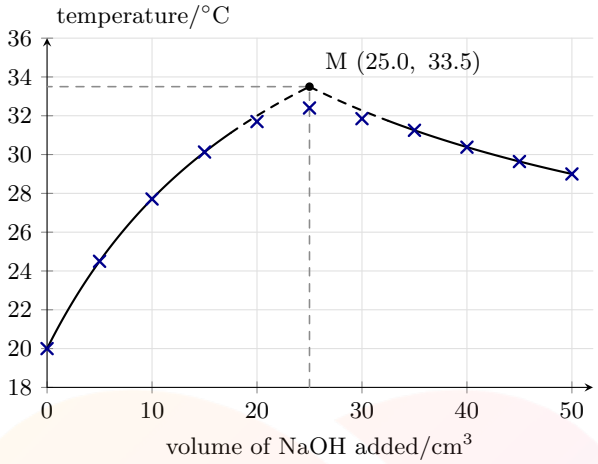
**A** 6 **B** 8 **C** 9

**D** 13 **E** 11 **F** 3



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**16.** In a thermometric titration, 2.00 mol $dm^{-3}$ sodium hydroxide solution is added in small portions from a burette to 25.0 $cm^3$ of dilute sulfuric acid in an insulated cup. The mixture is stirred and its temperature recorded after each addition. Both solutions start at 20.0 °C. The curves drawn through the two sets of points are extended (dashed) to meet at M, which gives the volume of alkali needed for neutralisation and the temperature that would have been reached with no heat loss.




The experiment is repeated using 50.0 $cm^3$ of the same sulfuric acid and sodium hydroxide solution of concentration 4.00 mol $dm^{-3}$, again with both solutions starting at 20.0 °C.

Assume that there is no heat loss, that every solution and mixture has density 1.00 g $cm^{-3}$ and specific heat capacity 4.2 J $g^{-1}$ $K^{-1}$, and that the same amount of energy is released for every mole of water formed. What is the highest temperature reached in the repeat experiment?

**A** 29.0 °C **B** 36.5 °C **C** 47.0 °C

**D** 38.0 °C **E** 30.8 °C **F** 33.5 °C

**17.** In acidic solution, vanadium forms four coloured ions: $VO_2^+$ (yellow), $VO^{2+}$ (blue), $V^{3+}$ (green) and $V^{2+}$ (violet). Whenever vanadium ions whose oxidation states differ by more than one are present together, they react with one another until all the vanadium is in a single oxidation state, or in two oxidation states that differ by one.

A student dissolves 3.51 g of ammonium vanadate, $NH_4VO_3$, in excess dilute sulfuric acid, which converts all the vanadium into $VO_2^+$ ions without changing its oxidation state. She then adds a solution containing 0.010 mol of $V^{2+}$ ions, and finally 0.65 g of zinc powder. All the zinc dissolves, forming $Zn^{2+}$ ions, and the zinc reacts only with vanadium species. No other reactions take place (in particular, there is no oxidation by air).

Which of the following describes all the vanadium ions present at the end?

Relative atomic masses: H = 1, N = 14, O = 16, V = 51, Zn = 65.

**A** 0.010 mol of $VO_2^+$ and 0.030 mol of $VO^{2+}$

**B** 0.030 mol of $VO^{2+}$ and 0.010 mol of $V^{3+}$

**C** 0.020 mol of $VO^{2+}$ and 0.020 mol of $V^{3+}$

**D** 0.040 mol of $VO^{2+}$ only

**E** 0.010 mol of $VO_2^+$, 0.020 mol of $VO^{2+}$ and 0.010 mol of $V^{2+}$

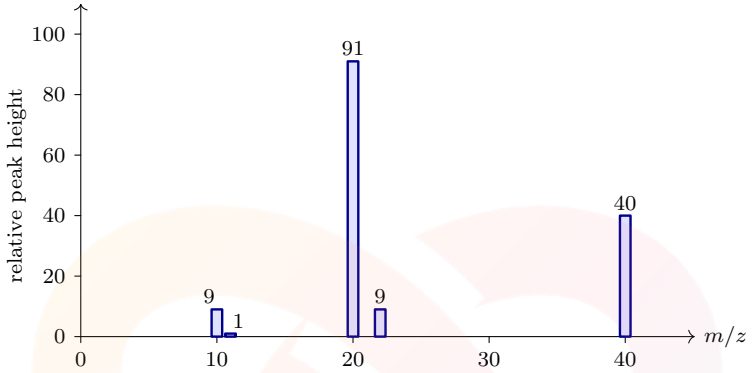
**F** 0.010 mol of $VO^{2+}$ and 0.030 mol of $V^{3+}$



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**18.** In a mass spectrometer, atoms are ionised and the ions are separated according to their mass-to-charge ratio, $m/z$, where $m$ is the relative mass of the ion and $z$ is the size of its charge. So a 1+ ion of relative mass 40 appears at $m/z = 40$, but a 2+ ion of the same mass appears at $m/z = 20$. Take the relative mass of each ion to be the mass number of its atom.

A sample of gas contains only neon and argon. Take neon to consist only of $^{20}$Ne and $^{22}$Ne, and argon to consist only of $^{40}$Ar. Each atom that is ionised becomes either a 1+ ion or a 2+ ion. The fraction of neon atoms that become 2+ ions is the same for both neon isotopes, but it need not be the same for argon. Assume that the same proportion of the atoms of each element is ionised and detected, and that the height of each peak is proportional to the number of ions arriving, whatever their charge. The spectrum is shown below.




What is the ratio of the number of neon atoms to the number of argon atoms in the sample?

**A** 2 : 1 **B** 1 : 2 **C** 11 : 4

**D** 9 : 4 **E** 5 : 2 **F** 9 : 5

**19.** When heated in hydrogen, some reactive metals form white crystalline *hydrides*, which are ionic and contain the hydride ion, $H^-$. Metal X is one of the first 20 elements and is in Group 1 or Group 2. Its hydride reacts vigorously with water, and the only products are the hydroxide of X and hydrogen gas.

When 0.63 g of the hydride of X reacts completely with excess water, 720 cm$^3$ of hydrogen is produced, measured at room temperature and pressure.

Consider the following statements.

I. X is calcium.

II. A hydride ion has the same electron configuration as a helium atom.

III. Exactly half of the hydrogen atoms in the hydrogen gas produced come from the water.

IV. If the molten hydride of X were electrolysed using inert electrodes, hydrogen would be produced at the negative electrode.

Which of the statements are correct?

(Relative atomic masses: H = 1, Li = 7, Be = 9, Na = 23, Mg = 24, K = 39, Ca = 40. One mole of any gas occupies 24 dm$^3$ at room temperature and pressure.)

**A** II only

**B** I and II only

**C** I, II and III only



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**D** I, II and IV only

**E** II, III and IV only

**F** I and III only

**G** II and III only

**H** I, II, III and IV

**20.** A bottle of sodium hydroxide solution has been left unstoppered, and some of the alkali has reacted with carbon dioxide from the air:

$2NaOH(aq) + CO_2(g) \rightarrow Na_2CO_3(aq) + H_2O(l)$

A technician titrates 25.0 $cm^3$ of the solution with 0.100 mol $dm^{-3}$ hydrochloric acid, using two indicators in turn.

Phenolphthalein is used first. Its end point is reached when all of the hydroxide ions have been neutralised and all of the carbonate ions have been converted into hydrogencarbonate ions, $CO_3^{2-} + H^+ \rightarrow HCO_3^-$. This takes 22.5 $cm^3$ of the acid.

Methyl orange is then added and the titration is continued. Its end point is reached when all of the hydrogencarbonate ions have reacted, $HCO_3^- + H^+ \rightarrow H_2O + CO_2$. This takes a further 2.5 $cm^3$ of the acid.

What percentage of the sodium hydroxide originally in the bottle has reacted with carbon dioxide?

**A** 12.5%

**B** 11.1%

**C** 20%

**D** 10%

**E** 25%

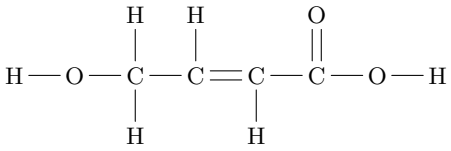
**F** 80%

**G** 22.2%



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**21.** Compound M has the displayed formula shown. It contains a C=C double bond, an OH group and a COOH group.



M can be polymerised in two different ways. Polymer A is the addition polymer formed using the C=C double bonds only. Polymer B is the condensation polymer (a polyester) formed when the OH group of each molecule reacts with the COOH group of the next, with loss of water; the C=C double bonds are unchanged. Treat both polymers as very long chains, so that the groups at the ends of the chains can be ignored.

I. When 1.02 g of A reacts with excess sodium, 240 $cm^3$ of hydrogen is produced, measured at room temperature and pressure (assume that every OH and COOH group reacts).

II. B undergoes addition reactions with bromine, but A does not.

III. Complete hydrolysis of either polymer gives M.

IV. Equal masses of A and B produce equal masses of carbon dioxide on complete combustion.

V. When 10.2 g of M is converted completely into B, 1.8 g of water is formed.

Which of the statements are correct?

Relative atomic masses: H = 1, C = 12, O = 16. One mole of any gas occupies 24 $dm^3$ at room temperature and pressure.

**A** I, II, III, IV and V

**B** II and V only

**C** II only

**D** I and II only

**E** I, II, IV and V only

**F** I, II and V only

**G** II, IV and V only

**H** I, II, III and V only

**22.** A supplier of photographic chemicals sells a white powder, used for making light-sensitive emulsions, that is a mixture of potassium bromide and potassium iodide only. A 4.04 g sample of the powder is dissolved in water, and exactly 240 $cm^3$ of chlorine, measured at room temperature and pressure, is bubbled through the solution until all of the chlorine has reacted.

The solution is then evaporated to dryness, and the solid is heated until its mass is constant. This drives off all of the bromine and iodine present, and leaves a residue of mass 2.68 g that contains only potassium salts. Assume that no other reactions take place.

What mass of potassium iodide was in the 4.04 g sample?

(Relative atomic masses: Cl = 35.5, K = 39, Br = 80, I = 127. One mole of any gas occupies 24 $dm^3$ at room temperature and pressure.)

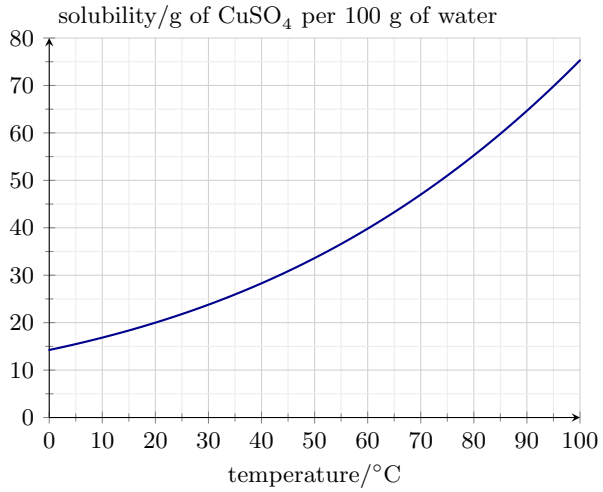
**A** 3.23 g **B** 2.35 g **C** 2.38 g

**D** 1.27 g **E** 1.66 g **F** 2.47 g



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**23.** The graph shows the solubility of copper(II) sulfate in water, measured as the mass of anhydrous $CuSO_4$ that dissolves in 100 g of water to give a saturated solution.




A student dissolves 100 g of blue copper(II) sulfate crystals, $CuSO_4 \cdot 5H_2O$, in 142 g of water at 80 °C. She then cools the solution slowly to 20 °C. Crystals of $CuSO_4 \cdot 5H_2O$ form, and the solution left is saturated. No water evaporates.

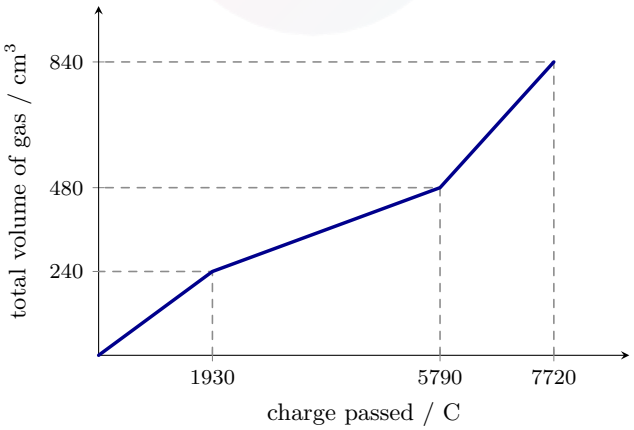
What mass of crystals forms?

(Relative formula masses: $CuSO_4 = 160$, $H_2O = 18$)

**A** 62.7 g **B** 55.6 g **C** 44.4 g

**D** 28.4 g **E** 71.6 g **F** 50.0 g

**24.** An aqueous solution contains only copper(II) sulfate and sodium chloride. It is electrolysed with inert electrodes using a direct current, and the total volume of gas given off at the two electrodes together is measured at room temperature and pressure. The graph shows this total volume against the charge passed.




Assume that at each electrode the product predicted by the usual rules for aqueous solutions forms for as long as the ion responsible remains in the solution, and that once that ion is used up the next product forms. In particular, chloride ions are discharged at the anode in preference to any other process while any remain. Assume also that all the gases are insoluble and that 96 500 C of charge transfers one mole of electrons.



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What masses of sodium chloride and copper(II) sulfate were dissolved in the solution?

Relative atomic masses: O = 16, Na = 23, S = 32, Cl = 35.5, Cu = 64. One mole of any gas occupies 24 $dm^3$ at room temperature and pressure.

**A** 3.51 g of NaCl and 1.60 g of $CuSO_4$

**B** 1.17 g of NaCl and 6.40 g of $CuSO_4$

**C** 0.585 g of NaCl and 4.80 g of $CuSO_4$

**D** 1.17 g of NaCl and 4.80 g of $CuSO_4$

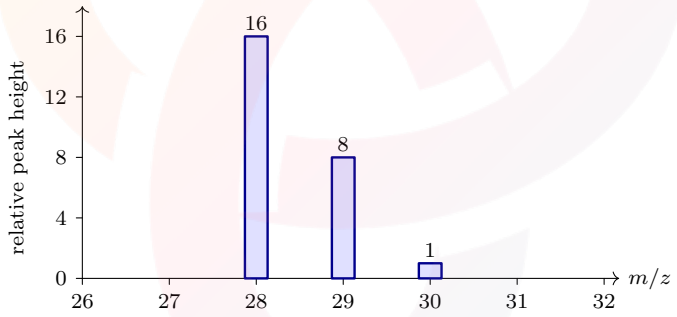
**E** 1.17 g of NaCl and 3.20 g of $CuSO_4$

**F** 1.17 g of NaCl and 9.60 g of $CuSO_4$

**25.** *Isotope dilution* is used to measure the mass of nitrogen in a sample of plant tissue. Take every nitrogen atom in the plant tissue to be $^{14}N$.

The analyst adds to the tissue a *spike* of 1.332 mg of ammonium sulfate, $(NH_4)_2SO_4$, made from enriched nitrogen in which 60% of the nitrogen atoms are $^{15}N$ and the rest are $^{14}N$. The spike and the tissue are mixed thoroughly, and all of the nitrogen they contain is converted into nitrogen gas, $N_2$, in which the nitrogen atoms are paired at random.

The mass spectrum of this nitrogen gas, showing only the peaks due to $N_2^+$ ions, is shown below.




What mass of nitrogen did the plant tissue contain?

(Take the relative masses of the atoms to be: H = 1, $^{14}N$ = 14, $^{15}N$ = 15, O = 16, S = 32)

**A** 0.28 mg

**B** 0.19 mg

**C** 0.56 mg

**D** 3.92 mg

**E** 0.67 mg

**F** 1.12 mg

**G** 0.84 mg

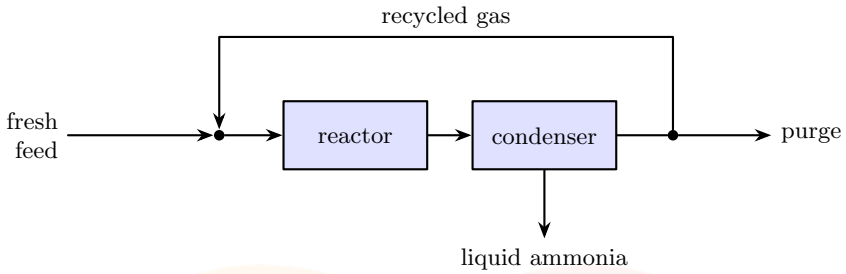


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**26.** In an ammonia plant, the fresh feed gas is nitrogen and hydrogen in the mole ratio 1 : 3, contaminated with a little argon. It joins a loop of recycled gas and passes through a reactor, where

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

Only 15% of the nitrogen and hydrogen entering the reactor is converted to ammonia on each pass. The gas leaving the reactor is cooled so that all of the ammonia condenses and is removed as a liquid. The remaining gas (unreacted nitrogen and hydrogen, and all of the argon) is recycled to the reactor, except that a small part of it is continuously vented as a purge, so that argon does not build up in the loop.




The plant runs at a steady state: the amount of every substance in every part of the plant stays constant. Every second, the fresh feed supplies 100 mol of $N_2$, 300 mol of $H_2$ and 4.0 mol of argon. The purge gas contains 1 mol of argon for every 5 mol of nitrogen and hydrogen combined. Argon does not react, and no gas dissolves in the liquid ammonia.

What mass of ammonia does the plant produce each second?

(Relative atomic masses: H = 1, N = 14)

**A** 3.23 kg **B** 1.62 kg **C** 3.26 kg

**D** 6.46 kg **E** 0.51 kg **F** 3.40 kg

**27.** An ester E has relative formula mass 158. Complete hydrolysis of E gives only a carboxylic acid and an alcohol, and these two products have the same relative formula mass. The acid contains one COOH group and the alcohol contains one OH group; neither contains a ring, a C=C double bond or any other functional group.

Considering every structure of E that fits this information, how many different carboxylic acids and how many different alcohols could the hydrolysis produce? Count structural isomers only, and ignore stereoisomerism.

Relative atomic masses: H = 1, C = 12, O = 16.

**A** 2 acids and 8 alcohols

**B** 2 acids and 3 alcohols

**C** 1 acid and 8 alcohols

**D** 2 acids and 6 alcohols

**E** 2 acids and 7 alcohols

**F** 4 acids and 4 alcohols

**G** 3 acids and 8 alcohols

*End of paper*



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